

Construction of a testable prototype

The final prototype iteration is clearly and fully explained and is constructed with enough detail to assure that objective data on all or nearly all design requirements could be determined; all attributes (sub-systems) of the unique solution that can be tested or modeled mathematically are addressed and a well-supported justification is provided for those that cannot be tested or modeled mathematically and thus require expert review.

When it came time to figure out if our design ideas would actually work, we decided to build a small-scale prototype. We used cardboard to stand in for steel, popsicle sticks to reinforce the structure, and string to represent the cables or chains. It wasn't fancy, but it gave us a way to visualize and test different frame configurations without spending a lot of money. Our main goal was to find a reinforcement strategy that could make the frame as strong as possible without driving up build costs. Along the way, we tried out several versions—experimenting with different support layouts, materials, and ways to fasten everything together.

- **Material**

In the end, we chose **Unistrut** as the material for the actual frame, and it wasn't a decision we made lightly. For one, Unistrut is incredibly strong thanks to its U-shaped design and high-grade steel construction—it gave us the structural integrity we needed without the complexity of welding custom beams. On top of that, the previous version of the project had already used Unistrut with 90-degree corner braces, so building on that foundation just made sense. It saved us time, money, and gave us a head start with parts and tools we were already familiar with. Using the cardboard allowed us to find a material that would meet the needs of the frame while being able to be strong enough to hold all the weight of these animals we engineered the frame to accommodate.

- Support layouts

The prototype also helped us create different layouts and different bracing formats as shown in the picture below (Popsicle Stick Frames).



Of the frames built, all had skin in the game for different reasons.

- The “Ribcage”
 - Used moderate amount of resources while being fortified enough to handle 36 marbles
- The “ZigZag”

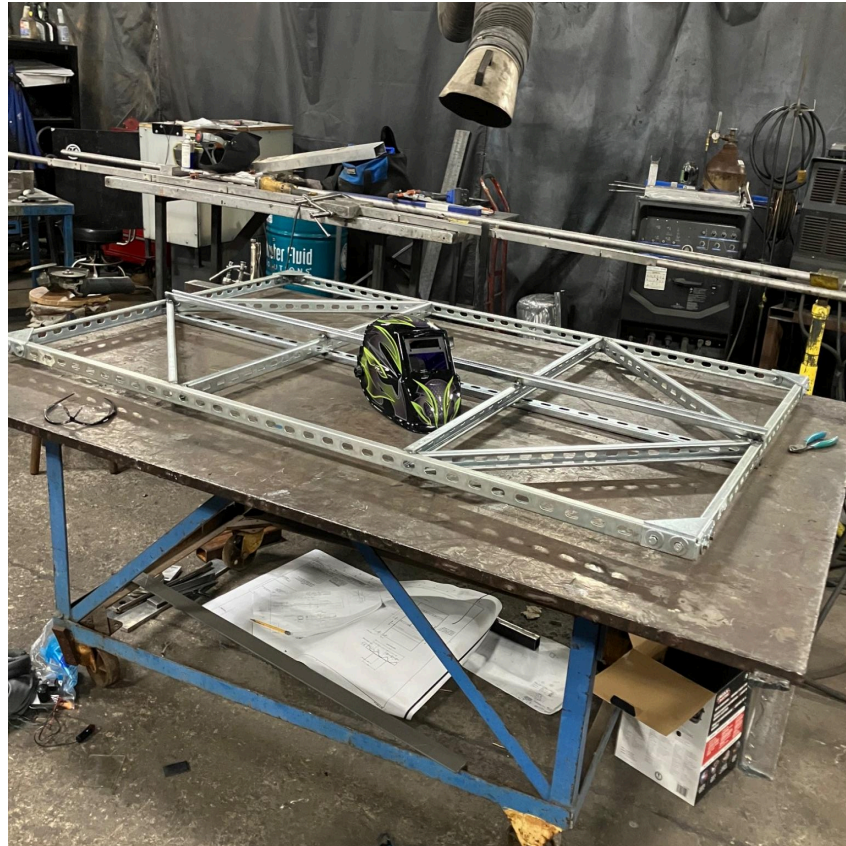
- Used a little over the required resources, yet stretched out weekpoints and had even strength around the perimeter of the frame
- 27 marbles
- The X
 - Used the least amount of resources, which was great for budget, but was by far the weakest
 - 18 marbles
- The “Coffin”
 - Used the most of resources, but made up for it in strength and fortification
 - 55 Marbles

As Shown by the picture, We opted for high resource for high strength and fortification, taking the Coffin Bracing and adding a secondary horizontal beam, along with another vertical beam going alongside the backside of the frame creating one of the strongest frames we could feasibly format.

- Facets

Although the Unistrut allowed us to use braces to connect the ends together to make the rectangle frame, we opted in for welding. Corbin, whose father owns a metalworking shop, has multiple years of experience welding. Corbin and Nic gutted the old frame of the connector braces from the preceding year, which Corbin then took to his shop,

welding the old frame together, while adding the bracing unistrut our group designed for the bracing.



With his spectacular welds, our frame was able to hold a total of 2,000 Lbs, in the form of a steel counterbalance, as shown below.

